

Class 5: 5G Is Not Just a Phone Network

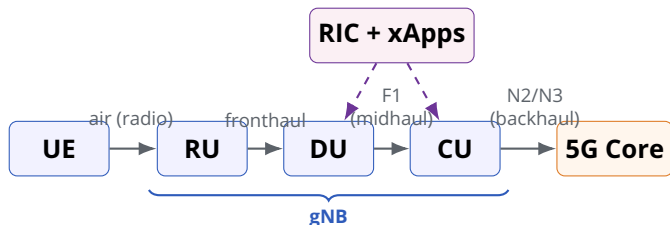
Seven Papers That Break the Architecture Diagram

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CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

The picture you spent two lectures building



Every box and line is exactly where you left it last class.

This picture is correct. What does it **silently assume**?

F1 = the DU-CU interface N2/N3 = the gNB-to-core interfaces

Five things the diagram never says out loud

Hidden in plain sight:

- 1 The base station stands **on the ground**.
- 2 The base station is **only a radio**.
- 3 Data is **transmitted first, computed later** (the wave carries bits; a computer does the math afterward).
- 4 The links **inside** the base station are trustworthy.
- 5 Attacking the network requires **standing up** a fake base station.

None of these are drawn. That is exactly what makes them assumptions.

By the end of today, all five will be broken — plus one so well hidden it is not even a box in the diagram.

Today's tour: four acts, seven papers

	The question it asks	Papers
Act I	Where does the base station <i>live</i> ?	2 papers, SIGCOMM'25
Act II	What lives <i>inside</i> the base station?	2 papers, SIGCOMM'25 3 papers + 1 cliffhanger
Act III	Who attacks it, and how?	NDSS'24 + USENIX Sec '19/'24/'25
Act IV	So what is the diagram, really?	synthesis — no new papers

SIGCOMM, NDSS, USENIX Security = the top venues where networking and security research is published

The goal is **not** to master seven papers today.

It is to leave feeling that **every box and every line can be reimaged.**

The course changes gears today

Classes 2–4

- “Here is how it works.”
- Standards, boxes, splits
- Answers with settled names

Class 5 onward

- “Which assumption breaks?”
- Papers, prototypes, attacks
- Questions nobody has settled yet

Q: If the diagram is correct, why spend a whole class attacking it?

A: Because **correct** is not the same as **finished**. The diagram tells you what was built; it stays silent about what was assumed. Research lives in that silence — and starting today, so does this course.

The tower left the ground

Classic Starlink needs a dish on your roof.

What if your ordinary phone talks straight to the satellite?

Not hypothetical: T-Mobile + Starlink have sold satellite texting on unmodified recent phones since July 2025.

Assumption: The base station stands on the ground

Broken by: a cell tower moving at about 7.5 km/s in LEO

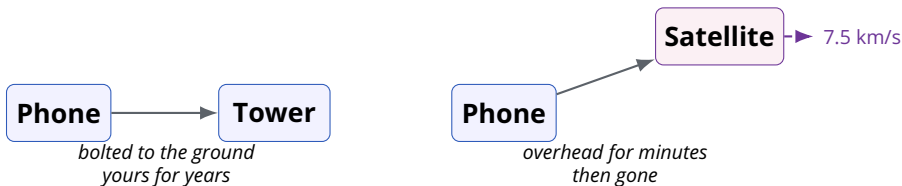
Direct-to-Cell Satellite Network without Satellite Navigation

Wei Liu, Yuanjie Li, Jingyi Lan, Hewu Li, et al. — Tsinghua University

ACM SIGCOMM 2025 [paper](#)

LEO = Low Earth Orbit D2C = direct-to-cell

Same phone. Very different tower.



Your base station now moves at 7.5 km/s.
Sixty seconds: shout out everything that breaks.

Keep your list. This lecture pays several of these off.

The twist: it secretly runs on GPS

Today's D2C networks lean on GNSS (GPS and friends).

- **Where you are** decides beam, country, and bill.
- **What time it is** decides when your radio may talk (uplink slots run on a shared clock).
- Both answers come from a different satellite system.

A ground tower knows where it is. A moving one has to keep asking. But GNSS can be **jammed**, **spoofed**, or simply **unavailable**.

Q: Satellite bars on your phone, but no GPS fix. What happens?

A: Field tests: denials, dropped access, wrong bills — **even with the D2C satellite overhead**. One satellite system's correctness hangs on another one.

GNSS = Global Navigation Satellite System (GPS, Galileo, BeiDou, ...)

beam = the patch of ground the satellite aims at uplink = the phone-to-satellite direction

The move: let the network navigate itself

SN2's idea: stop borrowing. Self-navigate.

- The D2C satellites themselves become the position source.
- If you can hear the satellite, you can locate by it.
- They live or die together — the authors call it **fate-sharing**.

Q: Doesn't the network need GPS-grade precision?

A: No. Beam, country, and bill need **good-enough** position, not meter-level.

▶ [SIGCOMM'25 talk — play 60–90 s](#)

What refusing to need GPS bought

Network availability

4.4–23.5×

higher than legacy
GNSS-dependent designs

Evaluated with commodity phones with 3GPP NTN support. Ranges span the scenarios tested.

NTN = Non-Terrestrial Network, 3GPP's name for satellite cellular

Access latency

1.9–12.3×

lower, same comparison

The fix was not better GPS.
It was deleting the hidden dependency on GPS.

Why is Netflix fast at 8 pm?

Q: Millions stream at once. Why doesn't it collapse?

A: Because the video rarely comes from far away. CDNs (Content Delivery Networks) cache popular content in servers **near you** — your city, sometimes your ISP.

Now route your Internet through space.

Every request climbs to a satellite, drops to a ground station, finds the cache on Earth — then climbs all the way back up.

Assumption: The cache sits on the ground, near the user

Broken by: putting the CDN on the satellites themselves

So well hidden it is not even a box in your diagram.

StarCDN: Moving Content Delivery Networks to Space

W. X. Zheng, A. Taneja, M. Masood, A. Sabnis, R. K. Sitaraman, D. Vasisht —
UIUC, UMass Amherst & Akamai

ACM SIGCOMM 2025 [paper](#)

► [SIGCOMM'25 talk — play 60–90 s](#)

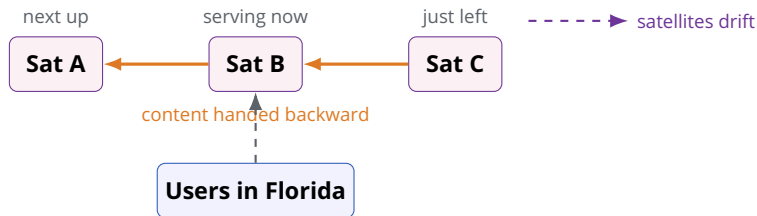
Your cache is doing 7.5 km/s

A LEO satellite laps the Earth in roughly 95 minutes.
The one over Florida right now is Florida's for only minutes.

Q: It spent those minutes caching what Floridians watch. Then what?

A: StarCDN hands the cached content **backward** through the constellation (the whole fleet) — against the satellites' motion — to the next satellite arriving overhead.

Satellites carry laser links to their neighbors — content rides those.



A useful way to picture it: the satellite moves on; the content stays behind. (Picture-level, not real geometry.)

What moving the cache to orbit bought

Ground-satellite bandwidth

~80%

less usage on the scarcest link

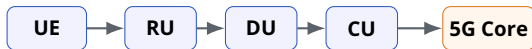
Trace-driven simulation using real-world Akamai CDN traces.

User-perceived latency

~2.5×

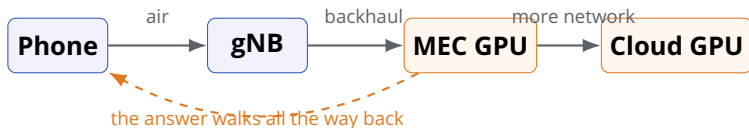
improvement over ground-based
CDN baselines

Act I score: the **tower** box and the **CDN** box both left the ground — and the systems got better.



Look at the rest of the architecture diagram.
Which boxes are on the ground only by habit?

The long way to the nearest GPU



Nearly every network-side “AI at the edge” demo still rides this path.

The gNB touches every byte — and **forwards all of them**.

MEC = Multi-access Edge Computing gNB = the 5G base station

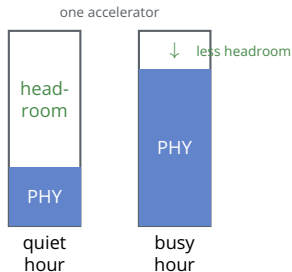
Assumption: The base station is only a radio

Broken by: a gNB that moonlights as an AI server

The accelerator was already in the building

Class 4, remembered:

- vRAN runs the PHY as software on servers.
- Those servers increasingly ship GPUs/NPUs for PHY math.
- PHY load rises and falls with traffic.
- The accelerator is **not always full**.



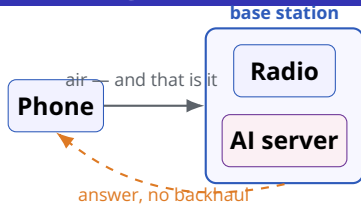
Illustrative shares, not measurements.

vRAN = virtualized RAN NPU = Neural Processing Unit

PHY = physical layer: decoding radio waves into bits — heavy number crunching

Spare accelerator + the best location in the network = a GPU nobody billed for.

AoRA: the gNB moonlights as an AI server



- Inference runs in containers on the PHY's spare headroom.
- Built on standard O-RAN interfaces — the ones you met in Class 4.
- Reported: transport latency cut over 30% vs MEC, 70% vs cloud.

transport latency = the network journey, not the model's compute time

AoRA: AI-on-RAN for Backhaul-free Edge Inference

S. Kholmatov, S. Cho, S. Chong, K. Lee — KAIST and Seoul National University
ACM SIGCOMM 2025 (short paper) [paper](#)

► [AoRA session recording \(SNU NXC\) — verify timestamp](#)

Busy hour: who gets the GPU?

Q: When the cell gets busy, who gets the GPU — the physical layer or the neural network?

A:

Busy hour: who gets the GPU?

Q: When the cell gets busy, who gets the GPU — the physical layer or the neural network?

A: The PHY, always. Miss its deadlines and the **cell itself** fails — no radio, no anything. AoRA plays it safe: when PHY load spikes, the AI steps aside and falls back to MEC or cloud.

The safe answer is not the **final** answer:

- PHY load high → AI gets less. PHY load low → AI gets more.
- Nobody has settled how to schedule that seam, second by second.

Communication-compute co-scheduling is young enough that a student in this room could move it.

Can the air itself be a computer?

Can the air itself be a computer?

Assumption: Data is transmitted first, computed later

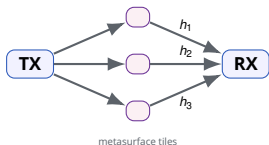
Broken by: a radio wave that arrives already half-computed

Same math, different department

What the channel does

$$y = \sum_i h_i x_i$$

Overlapping waves **add up**,
weighted by their paths.



each x_i = the signal copy riding one path;
the tiles set its weight h_i

Physics gives this away free.

A **programmable metasurface** lets software set the h_i .

Set $h_i = w_i$, and **propagation is the layer**.

Metasurface a flat panel of tiny reflectors whose response software can steer

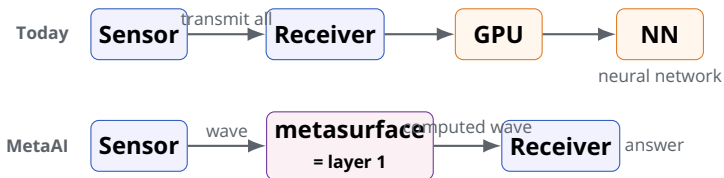
What a linear layer does

$$y = \sum_i w_i x_i$$

Inputs **add up**,
weighted by learned weights.

GPUs charge for this.

MetaAI: the wave arrives partially computed



Communication and computation stop being **sequential** steps.
They become the **same physical process**.

The receiver reads the summed wave as the layer's output — then finishes the (small) rest of the network.

Enabling Over-the-Air AI for Edge Computing via Metasurface-Driven Physical Neural Networks

C. Feng, S. Liang, C. Li, G. Zhao, B. Jing (Northwest Univ.), Y. Xie (Univ. at Buffalo), X. Chen (Northwest Univ.)

ACM SIGCOMM 2025 [paper](#)

What they built — honestly

Real hardware, modest brains:

- Prototypes at 2.4/5 GHz and 3.5 GHz — Wi-Fi and 5G territory.
- Average classification accuracy 82.8%, peaking at 89.8%.
- Scope today: **linear** layers, simple recognition tasks.
- This is a first step. It is not GPT-in-the-air.

Q: 82.8% would fail a Kaggle leaderboard. Why does SIGCOMM care?

A: Because the number is not the contribution — the **existence proof** is. A neural layer ran on real RF hardware, computed while the wave traveled — no GPU touched it. Leaderboards come later.

AoRA moved the AI **into** the box. MetaAI dissolved the box **into** the air.

Can we install antivirus on a cell tower?

Why did PCs get antivirus?

Not because they were computers.

Because they became **programmable platforms** anyone could ship software to — including defenders.

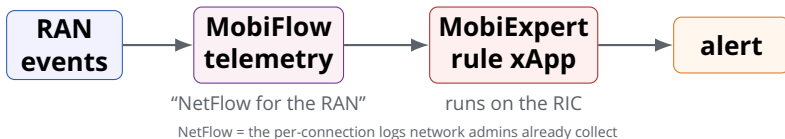
And the RAN?

For decades a base station ran one vendor's locked firmware.

Last class you learned O-RAN opened it: **RIC + xApps**.

If the tower is now a platform, can you install a security app on it?

Not deep learning — telemetry plus rules



A stream of fine-grained cellular telemetry feeds a rule engine that watches for known layer-3 (the signaling layer: attach, incoming-call alerts, handover messages) attack patterns. No model to train, no black box.

5G-Spector: An O-RAN Compliant Layer-3 Cellular Attack Detection Service

Haohuang Wen, Zhiqiang Lin (Ohio State); Porras, Yegneswaran, Gehani (SRI)
NDSS 2024 [paper](#)

What it caught, and what it cost

- Detects **7 classes** of layer-3 cellular attacks in real time.
- Shown to scale to **11 previously unknown attacks** by writing new rules.
- CPU overhead under **2%**; memory under **100 MB**.
- Won an **NDSS 2024 Distinguished Artifact Award**.

Q: So was the xApp box on our O-RAN slide just a PowerPoint concept?

A: No. Someone shipped a working security function inside it — and a committee reran it. The box on your diagram is a place code lives.

Attacking without building a tower

Assumption: Attacking the network requires **standing up** a fake base station.

Broken by: **Whispering in perfect sync with the real tower, slightly louder.**

A rogue base station is loud, obvious, and easy to look for — it is a whole tower that should not be there.

SigOver never builds one. It **synchronizes** to the legitimate LTE downlink and overwrites only the subframes it chooses (slices of the LTE frame, a millisecond each).

Think: two voices, identical timing — next slide.

You do not need a new tower to lie —
you need better timing.

LTE = the 4G radio standard downlink = the tower-to-phone direction



Jamming, spoofing, overshadowing

Jamming	You hear nothing — the channel is drowned out.
Spoofing (fake BS)	You switch towers — a new cell lures you over.
Overshadowing	You hear the attacker but believe it is your tower.

Two people speak to you with **identical timing**. Your ear keeps the louder one — and never notices there were two.

Scope: this is LTE (2019). It targets LTE **broadcast** messages, which are never integrity protected — before or after authentication. It is not a claim about 5G user traffic.

Hiding in plain signal

- A signal only **3 dB** (about twice the power) stronger wins.
- Success rate **98%** — vs **80%** for a fake base station.
- That fake base station needed a **35 dB** (about 3000×) edge.
- Timing is why 3 dB is enough: a time-aligned signal wins the receiver's capture.
- A fake tower has no such luck: it must out-shout cell selection.

capture = the radio locks onto the strongest aligned signal and treats the rest as noise
cell selection = how your phone picks which tower to camp on

Hiding in Plain Signal: Physical Signal Overshadowing Attack on LTE

Hoon Yang, Sangwook Bae, Mincheol Son, Hongil Kim, Song Min Kim, Yongdae Kim (KAIST)

USENIX Security 2019 [paper](#)

► [USENIX Sec'19 talk — play the demo, ~90 s](#)

What if the attacker never transmits a radio signal?

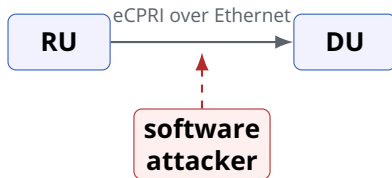
SigOver was intensely physical: a radio, microsecond timing, power measured in decibels.

Now turn the question inside out.

What could you break from software, without touching the air at all?

The answer is on a link you drew last class and never thought twice about.

The link inside the tower is now a network



Before: fronthaul was dedicated point-to-point fiber (CPRI) — effectively private wiring, physically trusted.

Disaggregated 5G: switched, packet-based Ethernet (eCPRI) — and to stay fast enough, nobody is required to check packets are genuine.

Framing: the fronthaul standards do not require integrity protection, citing performance overhead and assumed low risk. That assumption is what the paper tests.

RU = Radio Unit (antenna end) DU = Distributed Unit (baseband computer) eCPRI = the packet protocol between them

Assumption: The links inside the base station are trustworthy

Broken by: a software attacker on the fronthaul Ethernet

What software alone could do

On a commercial-grade, multi-cell testbed — two compromised cells, four users:

- User performance degraded by more than **80%**.
- Targeted users **prevented from ever attaching** to the cell.
- Signaling storms (floods of junk control messages) over **2500 msgs/min**.

On the Criticality of Integrity Protection in 5G Fronthaul Networks

Jiarong Xing (Rice); Yoo (Princeton); Foukas (Microsoft); Kim (UT Austin); Reiter (Duke)

USENIX Security 2024 [paper](#)

Q: The same opened seams let 5G-Spector in — and this attacker in. Safer, or more dangerous?

A: Both. Openness is a trade, not a verdict — which is why this is research, not settled fact.

The lineage did not die with LTE

SigOver's core idea — attack without a rogue tower — came back for 5G NR.

- Real-time sniffing of **pre-authentication 5G NR** traffic.
- Targeted downlink message injection, **no rogue base station**.
- **70–90%** injection success at up to **~20 m**.

5G NR = the 5G radio standard pre-authentication = messages sent before the phone proves its identity

Sni5Gect: A Practical Approach to Inject aNRchy into 5G NR

Shijie Luo, Matheus E. Garbelini, Sudipta Chattopadhyay, Jianying Zhou (SUTD)

USENIX Security 2025 [paper](#)

You will meet this paper again —
one of you presents it in the practicum.

Five assumptions walked in. None walked out.

	The move	Act
On the ground	Earth → orbit	I
Only a radio	GPU, AI, antivirus moved in	II-III
Transmit, then compute	the wave computes in flight	II
Inside links trusted	fronthaul is an attack surface	III
Fake base station	perfect timing instead	III

...and one assumption was never even a box: the cache sits on the ground (Act I).

One synthesis: **openness** → innovation **and** attack surface.

Seven papers and one teaser, one shared trick:
find the sentence the diagram never says.

How to read a paper like today's

- 1 Find the **assumption** it breaks.
Every paper today had exactly one.
- 2 Ask what it **deliberately does not solve**.
That silence is where the next paper lives.
- 3 Ask what **you** would build on top of it.
That question is the whole practicum.

Steps 1 and 2 need the paper. Step 3 needs **you**.

The 5G architecture diagram
is **not the answer.**

It is **the design space.**

Until today, this course taught you how it works.

From here, it asks what *you* would redesign.

The paper practicum starts soon — and Sni5Gect is already on the
list.